NEAR-INFRARED CLASSIFICATION OF VARIABLE STARS IN THE VVV SURVEY

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ABSTRACT

Identifying variable stars in the Galactic bulge is challenged by extreme extinction and an irregular sampling cadence, which complicates period searches due to aliasing. While near-infrared data from the VVV Survey mitigates extinction, it introduces a small signal-to-noise ratio. To address this, we developed a robust machine-learning pipeline based on the minimum entropy method for classification feature extraction. The classifier achieved 0.97 accuracy, with 0.95 and 0.98 recalls for variable and non-variable classes, ensuring high recovery with low contamination. This pipeline discovered ~10^3 unreported variable stars, laying the groundwork for VISTA's largest exclusive catalog of variable stars.

OBJECTIVE

To develop a robust integrated machine learning pipeline for stellar variability classification, targeting VVV Survey fields located in the Galactic bulge.

INTRODUCTION

The VVV Survey (Minniti et al., 2010) provides critical near-infrared data (specifically in the Ks-band) to explore the Galactic bulge. Previous extensive efforts, such as the VIVA-I catalog (Ferreira Lopes et al., 2020), suffer from high contamination (a 10:1 non-variable to variable ratio) due to extreme photometric noise, which frustrates classification strategies relying on standard variability and correlation indices. Furthermore, existing Machine Learning classifications often rely on numerous heuristic features lacking statistical discriminability in low Signal-to-Noise Ratio (SNR) regimes, yielding overly complex and inefficient models (Cabral et al., 2020).

To overcome these limitations, Information Theory offers a robust alternative. By maximizing phase-space order, Shannon entropy (Cincotta et al., 1995) uncovers variability patterns hidden in highly noisy signals, extracting features with deep physical and statistical interpretability. This theoretical framework is key to reliably identifying variables in regions characterized by severe stellar crowding and extreme extinction.

METODOLOGY

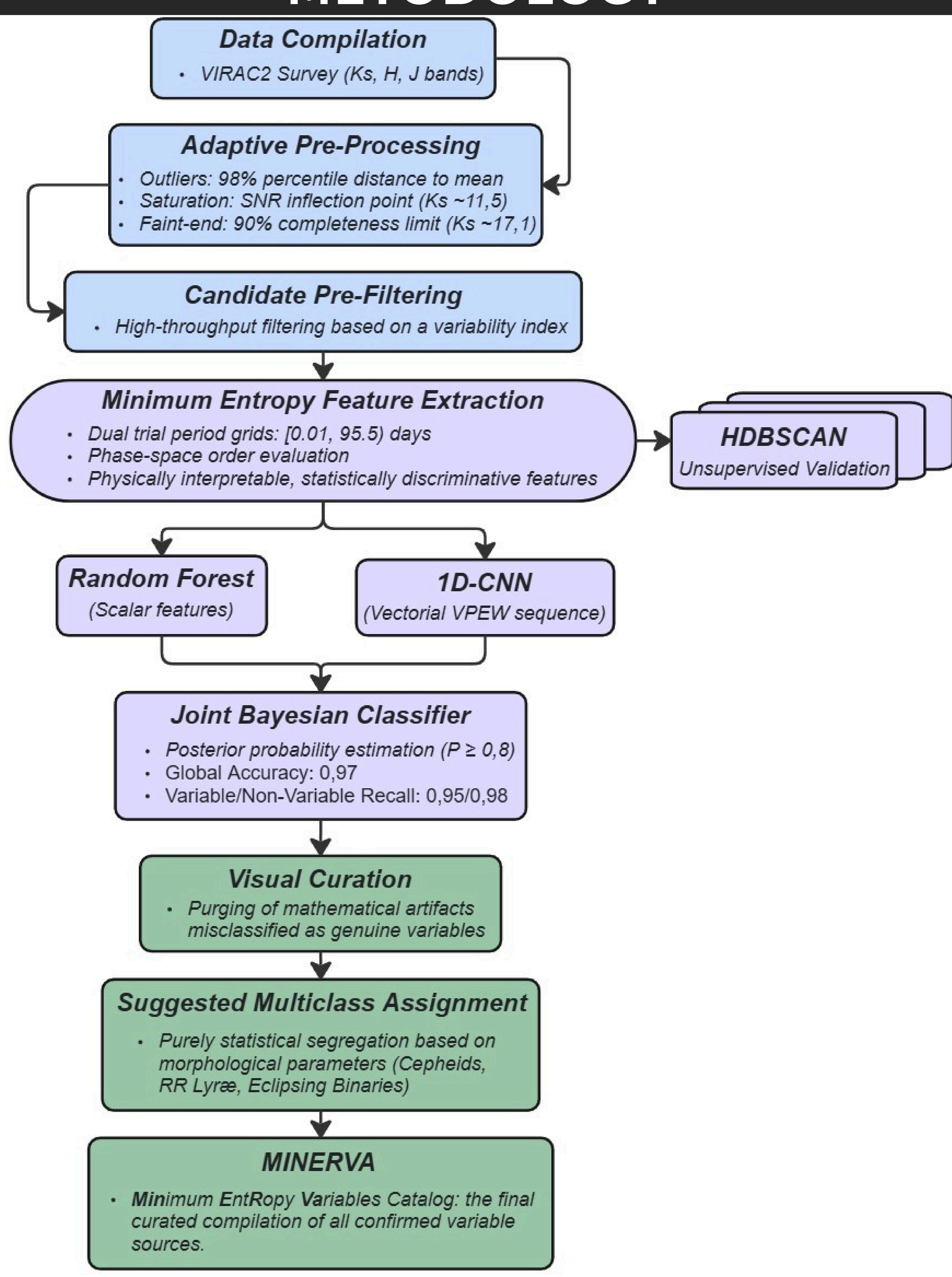

Figure 1. The MINERVA Extract, Transform, Load (ETL) classification pipeline.

RESULTS

Saturation & Faint-End Cutoffs

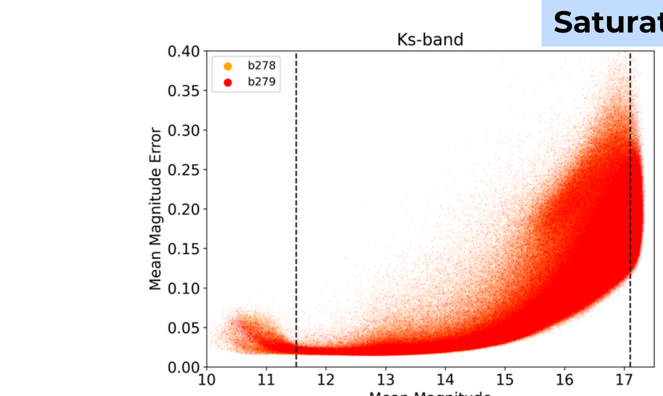

Figure 2. Mean magnitude error vs. mean magnitude. Dotted lines denote the applied saturation and faint-end thresholds.

Field	Saturated	Highly Reliable	Faint
b278	3.061	1.554.449	179.314
b279	3.051	1.306.177	146.249

Table 1. Distribution of sources according to their photometric regime.

Candidate Pre-Filtering

$$X^2_{var} = \frac{1}{N-1}\sum_{i=1}^{N}\frac{\left(m_i - \bar{m}\right)^2}{\sigma_i{}^2}$$

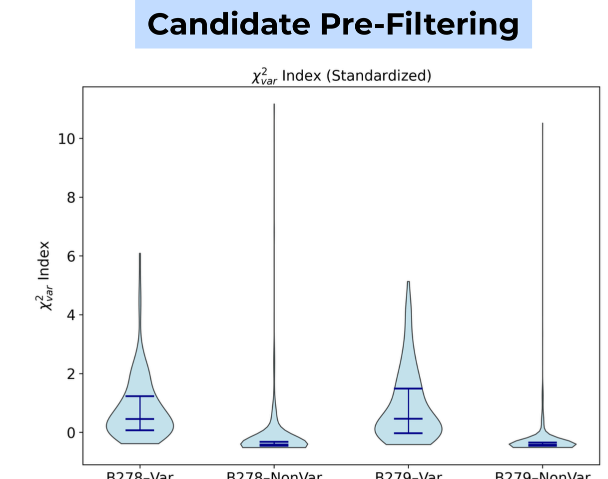

Figure 3. Distribution of the X^2_{var} index for variable and non-variable stars, demonstrating its strong discriminative power.

Field	Candidate Sources	Retained Fraction
b278	131.892	8,48%
b279	124.798	9,55%
Total	256.690	8,97%

Table 2. Pre-filtered candidate sources exhibiting significant variability ($X^2_{var} > 2$).

Minimum Entropy Feature Extraction

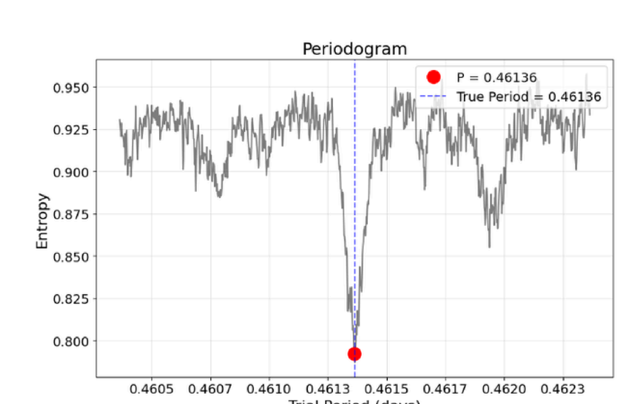

Figure 4. Shannon entropy periodogram. Deepest minima correspond to optimal phase-space organization periods.

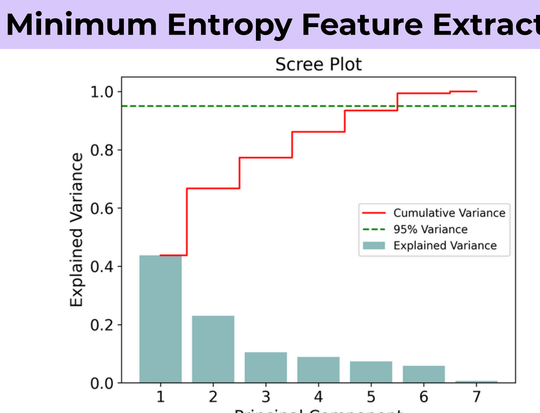

Figure 5. PCA scree plot. Six scalar features are required to explain >95% of the data variance.

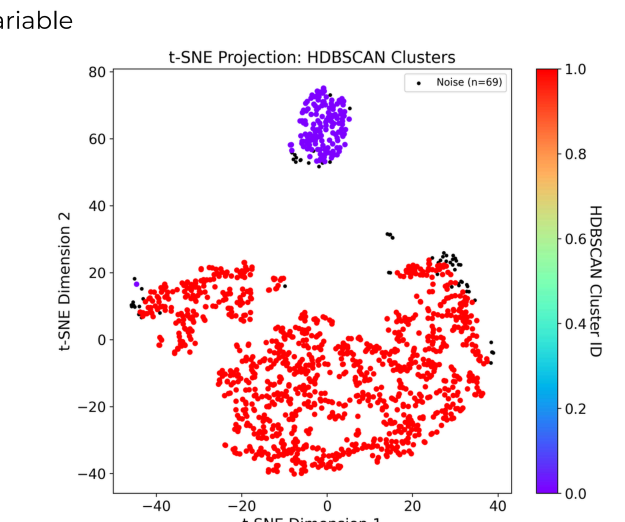

Figure 6. HDBSCAN t-SNE projection. Cluster 0 (violet) isolates VAR stars (99.3% purity); Cluster 1 (red) contains primarily NV stars (88.8%). Black points denote noise.

The MINERVA Catalog

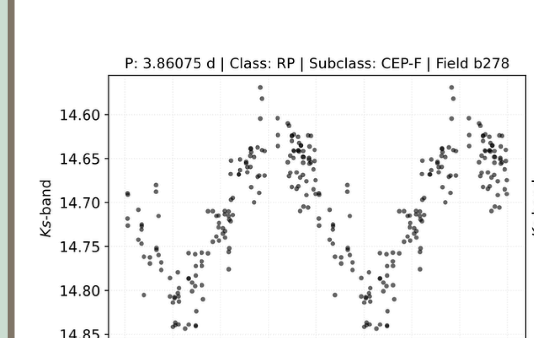

Figure 7. Example Cepheid phase-folded light curve.

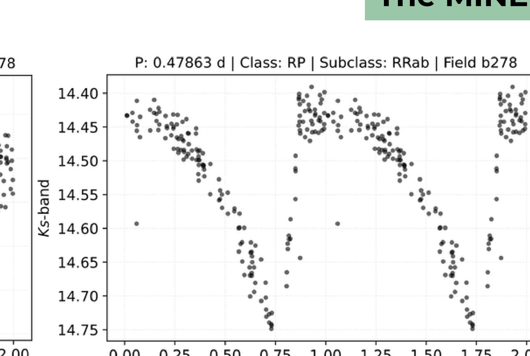

Figure 8. Example RR Lyrae phase-folded light curve.

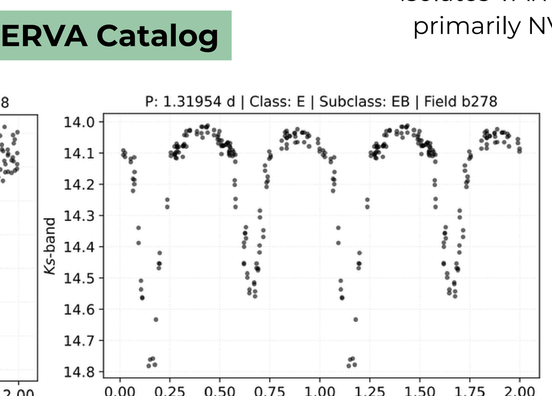

Figure 9. Example Eclipsing Binary phase-folded light curve.

Variability Class	Field b278	Field b279	Total
Cepheids	367	419	786
RR Lyræ	301	272	573
Eclipsing Binaries	121	113	234
UVAR	311	343	654
Total Subset	**1.100**	**1.147**	**2.247**

Table 3. MINERVA catalog composition by variability class.

DISCUSSION

- **Breaking the Noise Barrier:** By evaluating phase-space coherence, the minimum entropy method extracts highly discriminative features that break through low SNR and extreme photometric noise. To guarantee minimum model complexity and avoid redundancy, we ensured minimal linear correlation among these features. Furthermore, Kruskal–Wallis testing confirmed that each extracted feature is statistically significant for distinguishing between the Variable and Non-Variable classes.
- **Unprecedented Efficiency:** Our Joint Bayesian Classifier offers a highly competitive alternative to previous pioneering VVV efforts. While earlier ML applications reported an average recall of 0.48 for RR Lyrae (Cabral et al., 2020), our pipeline achieved a 0.97 overall accuracy and a 0.95 recall across all variable classes. Crucially, it achieved a 0.98 recall for non-variables. This exceptional rejection capability provides a significant improvement over the 10:1 contamination ratio reported in the foundational VIVA-I catalog (Ferreira Lopes et al., 2020), allowing for a highly purified sample.
- **Survey-Wide Expansion:** Extrapolating a classification pipeline across the VISTA project is a major challenge because observational cadences, the number of observed epochs, and stellar densities vary drastically across regions. Our framework's robustness stems from an adaptive pre-processing strategy that calculates independent, reliable photometric thresholds for each field—effectively excluding saturated and extremely faint stellar populations. By avoiding rigid global parameters and adapting strictly to local observing conditions, this methodology is perfectly poised for application across the diverse environments of the Galactic bulge and southern disk.

CONCLUSIONS

- Maximizing phase-space order via Shannon entropy provides a superior, physically interpretable foundation for classifying stellar variability in heavily obscured, noise-dominated environments.
- The application of this pipeline to fields b278 and b279 yielded a highly curated dataset of 2,247 confirmed, previously unreported variables, each parameterized with precise periods and suggested multi-class labels.
- The adaptive pre-processing and statistical robustness of this framework guarantee its scalability across the entire 348-tile VVV footprint. This extrapolation projects a definitive yield of 10^5 confirmed variables, stablishing MINERVA as VISTA's premier, high-confidence resource for Galactic structure and stellar evolution studies.

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SIMULATE & EXPLORE THE DATA